Uniform mesh refinement underestimates the frequencies of the DeviceLayout.jl transmon: an independent finite-element cross-check and re-optimization

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Abstract

The SingleTransmon example of DeviceLayout.jl publishes qubit and resonator frequencies for an initial and an optimized design, computed with the Palace solver on one mesh. We find those frequencies discretization-limited. Palace reproduces them within 0.7%, and geode-fem, an independent solver, cross-checks Palace: on shared meshes the two agree to 2×10^{-4}% on the qubit and 0.03% on the resonator. All frequencies are linear eigenmodes. Adaptive Palace runs, still rising, put the qubit frequencies 7.0% and 8.0% above their published values without extrapolation. On the initial design, a two-sided bound on the island capacitance, read through a circuit model, caps the qubit at 4.477 GHz (+8.1%). Uniform refinement converges slowly, because the field is singular at the metal edges. Extrapolated assuming error $\propto h$, the fastest rate those edges allow on uniform meshes, two refinement families put the qubit and resonator frequencies 7.3–9.6% and 2.7–3.3% higher; these estimates lean low and rest on Palace's finer levels. At these frequencies the 3.0/4.0 GHz design needs a capacitor 19–20% longer than published. Newton's method in DeviceLayout's variables meets the example's criterion in 3 or 4 Palace runs with any of three Jacobians, against 10 to 12 for PRIMA; geode-fem's shape derivative gives no advantage here.

1 Introduction

A transmon qubit is an electrical oscillator: a metal island, whose capacitance to its surroundings is the oscillator's capacitor, connected to ground through a Josephson junction, which acts as its inductor. Its frequency, which we call the *qubit frequency*, is the linear eigenfrequency of the junction's (plasma) mode.[1] It is set mostly by that capacitance, which a three-dimensional field solve computes, and the readout resonator coupled to it is sized the same way. Designers lengthen or shorten the island and the resonator until the computed frequencies sit on their targets, so if a solve reads a frequency 8% low, the design it steers to has, in the same device model, a frequency about 8% above its target.

DeviceLayout.jl (AWS Center for Quantum Computing, 2025a) builds such a layout and its mesh, and Palace (Palace developers, AWS Center for Quantum Computing, 2023), an MFEM-based finite-element solver (Anderson et al., 2021), computes the eigenmodes; tools such as QDesignOptimizer (Eriksson et al., 2025) steer the same loop with analytic models. DeviceLayout's SingleTransmon

[1] Not the transmon's 0–1 transition f_{01}, which lies below the plasma frequency by about the charging energy E_C/h (Section 3.3).

example (AWS Center for Quantum Computing, 2025b; Peairs and Carson, 2025) runs the loop end to end. It publishes an initial transmon-plus-resonator design at 4.14 and 5.591 GHz and a design optimized for targets of 3.0 and 4.0 GHz, at 3.005 and 3.982 GHz, found with a derivative-free optimizer in nine Palace runs. Like every frequency here, these are linear eigenfrequencies.

Those frequencies come from one discretization, and the example's authors say what that implies. They run Palace at second order "to reduce sensitivity to the discretization" and note that this "does not mean that the eigenfrequencies found by Palace will have converged to their true values" (AWS Center for Quantum Computing, 2025b). This paper measures how far from converged the published frequencies are. Our main result is that the published frequencies are discretization-limited. Palace's adaptive refinement puts the two qubit frequencies 7.0% and 8.0% above their published values without extrapolation, and its uniform ladders, in two refinement families (two ways of subdividing the same starting mesh), extrapolate to 7.3–9.6% above them; every uniform level we could afford lies below those estimates. geode-fem reproduces every level of those ladders it can reach, the two coarsest meshes with linear elements, to 10^{-4}% on the qubit, and a two-sided bound on the island capacitance, read through a circuit model, is consistent with that shift and caps the initial qubit under that model. The extrapolation rule was chosen after that capacitance route had been computed, so their agreement is a consistency check, not a blind prediction. Three kinds of interval appear, and Table 6 sorts every result by them: a *bound* holds by theorem and is given only for the island capacitance; a *cap* is a frequency limit that follows from a bound only under the circuit model; and a *band* is the range of an extrapolation, which is neither.

The second code is GEODE-FEM ("geode-fem"), a Rust finite-element solver on the Burn tensor framework (Tracel AI, 2026). It solves the same $H(\mathrm{curl})$ eigenproblem as Palace with a separate implementation, and it computes design derivatives, among them the shape derivative of an eigenfrequency. Given the same input mesh, it reproduces Palace's modes (Section 4), and on the $h/2$ meshes of Section 5 the two codes solve the same discrete problem apart from the junction term. This gives an independent same-mesh cross-check of the Palace/DeviceLayout.jl transmon stack, complementing the earlier checks of Palace by SQDMetal (Sommers et al., 2025) and Ye et al. (2025). Section 9 explains why this agreement places the shift in the mesh, not in either solver or its arithmetic.

Our contributions are:

- **A refinement study of a published design example**, which puts its qubit frequencies 7.3–9.6% above their published values, with a second refinement family and adaptive runs on both designs checking the qubit estimates; the finer levels and the extrapolations are Palace's (Section 5).

- **Two-sided capacitance bounds with the sapphire tensor.** An equilibrated-flux lower bound and conforming upper bounds bracket the island capacitance C_{isl} of the unmodified device, without the junction capacitance, in $[79.56, 79.77]$ fF with the box walls held at ground, as in the eigenproblem; under a two-node circuit model the lower bound caps the initial qubit at 4.477 GHz (Section 6).

- **A re-optimized design, with controls.** At the extrapolated frequencies the 3.0/4.0 GHz design needs $c \approx 1451$–$1466\,\mu$m against the published $1217.678\,\mu$m. Against the example's own objective, Newton's method in DeviceLayout's variables needs 3 or 4 Palace runs with any of three Jacobians and PRIMA 10 to 12. geode-fem's shape derivative is exact but gives no advantage on this two-parameter problem; its one-time calibration of the resonator model replaces one Palace probe run (Sections 7 and 9).

- **Same-input-mesh agreement with Palace**, with the 17.49 GHz mode that both codes share traced to the port model (Section 4).

2 Related Work

Cross-checks of Palace. SQDMetal (Sommers et al., 2025) cross-checks a Palace workflow against COMSOL and Ansys with mesh-convergence studies, and Ye et al. (2025) compare a Palace workflow with cryogenic measurement, matching resonator frequencies to within 0.3% with a mesh refined at the CPW gaps and fourth-order elements, a setting their convergence study puts within ±0.3% of extrapolated values. Both cross-checks use models built independently for each tool. Our comparison instead holds the input mesh and the junction model fixed and changes the solver implementation, and it adds a refinement study of a published design: code-to-code comparison and solution verification (Oberkampf and Roy, 2010), not validation against measurement. As of October 2026, neither Palace nor DeviceLayout.jl has a methods publication; we cite their repositories and documentation.

Design sensitivities and qubit-design optimization. Adjoint sensitivity analysis of frequency-domain electromagnetic solvers is an established field (Nikolova et al., 2004), including shape adjoints for Maxwell cavity eigenvalues (Akçelik et al., 2005); Palace exposes no derivatives. In qubit design, SQcircuit (Rajabzadeh et al., 2023) and SuperGrad (Wang et al., 2025) differentiate circuit Hamiltonians, SQuADDS (Shanto et al., 2024) looks designs up in what its authors call a "validated design database", and Eun et al. (2023) shape-optimize transmon pads with FEM and a derivative-free optimizer. DeviceLayout's example optimizes with PRIMA (Zhang, 2023), a reference implementation of Powell's model-based derivative-free methods; with bounds only it runs BOBYQA (Powell, 2009).

Two-sided capacitance bounds and circuit models. Bracketing a capacitance between complementary energy principles is classical (Specogna, 2011). Our lower bound uses the hypercircle idea of Prager and Synge (1947), which turns any field that nearly balances the charges into a lower bound once its imbalance is paid for, with Raviart–Thomas fluxes (Raviart and Thomas, 1977) equilibrated patch by patch (Braess and Schöberl, 2007; Ern and Vohralík, 2015). The method is not new; we apply it to an island capacitance with a floating conductor on a production qubit layout. A web search in October 2026 of the complementary-bound and qubit-design literature (queries such as "guaranteed capacitance bounds qubit") found no two-sided capacitance bound in the qubit-design literature. The two-node circuit of Section 3.3 is a minimal instance of the quasi-lumped route (Minev et al., 2021b), which partitions a device into lumped or quasi-distributed cells, simulates each and reduces the composite to coupled subsystems; the energy-participation route (Minev et al., 2021a) instead quantizes the distributed eigenmodes.

3 Solvers, Device and Models

3.1 geode-fem

geode-fem assembles first-order Nédélec (Nédélec, 1980) and scalar P1/P2 element operators as batched $[n_{\text{elem}}, k, k]$ tensor operations on Burn. Its eigensolver is shift-invert Lanczos (Lehoucq et al., 1998) on a real symmetric pencil, factorized with faer (Kazdadi, 2026). The two codes share the formulation, the lumped junction port, the materials and the input mesh, and geode-fem follows a specification of Palace's operators (Section 8); assembly, the eigensolver and the linear algebra are separate implementations.

The eigenfrequency derivative used in Section 7 needs no adjoint solve, because the pencil is symmetric: for a simple eigenpair (λ, x) with $x^{\mathsf{T}} M x = 1$,

$$\frac{d\lambda}{d\theta} \;=\; x^{\mathsf{T}}\Big(\frac{\partial K}{\partial \theta} - \lambda\,\frac{\partial M}{\partial \theta}\Big)x, \tag{1}$$

the classical eigenvalue derivative of a symmetric generalized eigenproblem (Fox and Kapoor, 1968). Here K and M include the junction terms of Eq. (2); the junction does not move under the capacitor-length motion of Section 7, so those terms do not contribute. geode-fem evaluates $\partial K/\partial\theta$ and $\partial M/\partial\theta$ along a node velocity $\partial X/\partial\theta$ with a hand-written forward-mode dual-number twin of its Nédélec kernel (tested against it to 10^{-10}); Burn's autodiff tape is not used.

3.2 Device, meshes and materials

The device is DeviceLayout.jl v1.15.0's SingleTransmon example (AWS Center for Quantum Computing, 2025a): a transmon (Koch et al., 2007) with a meander readout resonator and a feedline on sapphire, in a vacuum box (Figure 1). Its two design parameters are the transmon's `capacitor_length` c, the length of the island, and the resonator's `total_length` L, the length of the meander (Figure 1a); the published initial design has $c = 620\,\mu$m and $L = 5000\,\mu$m. The sapphire permittivity is $\mathrm{diag}(9.3, 9.3, 11.5)$ with the optic axis normal to the chip; the configuration rotates the material axes about that normal, which leaves a uniaxial tensor unchanged, so the lab tensor is diagonal. Where a scalar model is used it is the trace average $\bar{\varepsilon} = 10.03$. PEC (perfect-conductor) conditions hold on the metal sheets and on the box walls, which therefore sit at ground potential; the readout ports are left open in every eigenproblem except the DeviceLayout-configuration runs of Section 5.

Table 1 lists the meshes. Element order p is the polynomial degree of the finite elements ($p = 1$: linear, $p = 2$: quadratic); *straight-sided* and *curved* describe the geometry of the mesh, whose element faces are flat or follow curved surfaces. The *fixture* is a straight-sided mesh of the initial geometry written by an earlier DeviceLayout.jl v1.15.0 driver, `reference/julia/generate_transmon_fixture.jl`, pinned by SHA-256; its levels are h_F, $h_F/2$ and $h_F/4$. Refining it leaves its polygonal surfaces unchanged, which is what the capacitance bounds of Section 6 need. For the cross-check and the re-optimization, DeviceLayout meshes each geometry with the two parameters exposed, straight-sided at level h for the ladders and curved for the runs that reproduce its discretization; curving the faces changes the qubit by at most 0.0001% and the resonator by about 0.05%. On the initial geometry h_F and h give the same Palace qubit (3.4527 against 3.4520 GHz, 0.02%). A refinement *family* is one way of subdividing h uniformly: family a with gmsh, family b with Palace.

3.3 Junction and circuit models

The Josephson junction is a linear lumped element, $L_J = 14.860\,$nH in parallel with $C_J = 5.5\,$fF, on the four-triangle sheet Γ of Figure 1b, as in DeviceLayout's Palace configuration. With port length ℓ and width w the uniform lumped-port construction adds

$$K_{\mathrm{port}} = \frac{\ell}{w\,L_J}\,S_\Gamma \qquad \text{and} \qquad M_{\mathrm{port}} = \frac{C_J\,\ell}{w}\,S_\Gamma \tag{2}$$

to the pencil $(K + K_{\mathrm{port}})x = \omega^2(M + M_{\mathrm{port}})x$, where S_Γ is the isotropic surface matrix on Γ. Palace's `LumpedPort` adds the same terms with the same scalings. geode-fem's rank-1 port-voltage shunt replaces S_Γ by $b\,b^{\mathsf{T}}$, with $b_i = (1/w)\int_\Gamma N_i \cdot \hat{e}\,dS$ the port-voltage functional; on the fixture it moves the qubit by 6×10^{-5} %, keeps the other physical modes to four decimals, removes the artifact mode of Section 4, and is the junction model of every geode-fem eigensolve in Sections 5 and 7.

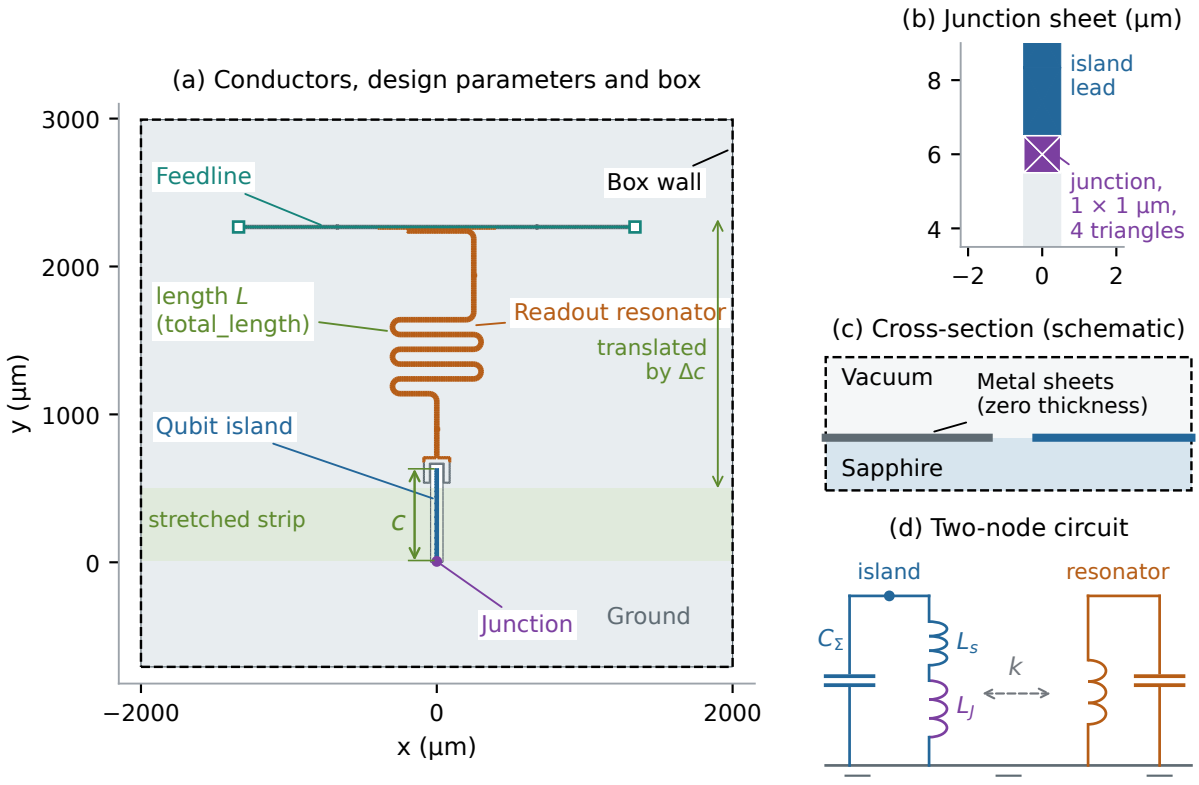


Figure 1: Benchmark geometry, drawn from the SHA-256-pinned MSH 4.1 input mesh of the initial design (the fixture; 22,684 nodes, 133,314 tetrahedra). (a) Top view of the three metal conductors of Section 6: the feedline (teal), the qubit island (blue, as in (b) and (d)), and the ground plane (gray) with the meander resonator (its outline orange), whose length is L; the junction (purple, as in (b) and as L_J in (d)) joins the island to ground, squares mark the two readout ports, and the dashed outline is the box wall. Each label has the colour of the object it names; the design parameters are annotated in green. c is the island length. When c changes, the shaded strip stretches and everything above it moves up by Δc (Section 7; bands from `fd_initial.toml`). (b) The $1 \times 1\,\mu$m junction `lumped_element` sheet with its four-triangle triangulation. (c) Material cross-section with the box walls (dashed), schematic and not to scale. (d) The two-node circuit of Section 3.3: the island's capacitance C_Σ across the junction inductance L_J and the series inductance L_s of its current path, coupled with coefficient k to the resonator.

The electrostatic model splits the metal into three conductors: ground with the resonator, the floating feedline, and the island. Its island capacitance with the feedline floating and neutral, $C_{\text{isl}} = C_{ii} - C_{if}^2/C_{ff}$, excludes the junction. In the convention of Koch et al. (2007) the transmon's total capacitance is $C_\Sigma = C_{\text{isl}} + C_J$, and it sets the charging energy $E_C = e^2/2C_\Sigma$. A two-node circuit (Figure 1d) connects it to the eigenproblem: a bare qubit $1/(2\pi\sqrt{(L_J + L_s)C_\Sigma})$, with L_s the series inductance of the junction's current path, coupled with coefficient k to the resonator. *Bare* is the qubit's own LC frequency; *dressed* is its frequency after the coupling to the resonator pushes it, which is what the eigenproblem computes. The same circuit is used both ways: forward from a bound on C_Σ to a dressed qubit, and backward from a dressed pair to C_Σ. At $L_s = k = 0$ this is the plasma frequency $\sqrt{8E_JE_C}/h$, and the transmon's $f_{01} \approx \sqrt{8E_JE_C}/h - E_C/h$ lies below it. On the fixture L_s and k were fitted on three and two discretizations (`benchmarks/transmon_loop_inductance/`, `benchmarks/transmon_coupling/`); extrapolated, they give fitted ranges of 0.113–0.125 nH and 0.0227–0.0310, which are estimates.

Table 1: The meshes. "Initial" and "final" are the two published designs; "step 2" is the re-optimized design of Section 7.

Mesh	Designs	How made	Levels	Used for
Fixture	initial	earlier DeviceLayout driver, straight-sided; refined uniformly, or adaptively by geode-fem	h_F, $h_F/2$, $h_F/4$; adapted	same-mesh agreement (§4); capacitance bounds and cap (§6)
DeviceLayout h	every design	DeviceLayout, straight-sided	h	ladders (§5); geode-fem solves and derivatives (§7)
Family a	every design	gmsh splits each tetrahedron of h into 8	$h/2$, $h/4$	ladders and estimates
Family b	initial, final, step 2	Palace's uniform refinement of h	$h/2$, $h/4$	a second estimate
Adaptive	initial, final	Palace adaptive refinement from h at $p = 2$	passes 1–5	check of the qubit estimates
DeviceLayout's discretization	every design	DeviceLayout, curved; Palace at $p = 2$	one	reproduction; the Newton objective F (§7)

4 Same-Input-Mesh Agreement With Palace

Both codes read the fixture with the same materials and junction model at $p = 1$ (Table 2). Palace's configuration schema defines `Target` as the frequency above which its eigensolver searches (Palace developers, AWS Center for Quantum Computing, 2023), so the six modes above 4.5 GHz come from a run with `Target` = 4.5 GHz and the qubit from one with `Target` = 2 GHz. All six agree to $\leq 0.033\%$, and the qubit mode to 5×10^{-5} %, with port participation 0.9942 in geode-fem and port EPR 0.99424 in Palace, the energy-participation ratio of the junction, the fraction of the mode's inductive energy stored in it (Minev et al., 2021a). Both codes use the isotropic sheet term of Eq. (2) here.

The two codes read the same input mesh but do not solve the same unknowns. Palace cracks each metal sheet into two sides by duplicating its vertices, which leaves the PEC problem unchanged, and where a sheet is one element wide, almost all on the resonator and feedline sheets, it also bisects the elements on both sides ("local bisection for under-resolved interior boundaries" in its log; `palace/utils/geodata.cpp`). Only the bisection changes the unknowns (Table 2).

Table 2: Eigenmode agreement from the identical input mesh (22,684 nodes, 133,314 tetrahedra), $p = 1$ in both solvers, junction as the lumped $L_J \parallel C_J$ shunt. Counted the same way, as all first-order Nédélec unknowns (edges) including those on PEC sheets, geode-fem's mesh has 156,863 (133,108 interior) and Palace's has 183,891, after it cracks the metal sheets and adds 4,913 elements on them by bisection, triggered by 1,007 edges, 993 of them on the resonator's ground plane and the feedline. Both codes are printed to geode-fem's archived three decimals (six for the qubit); Δ is the archived relative difference, computed from Palace's full-precision value. Sources: `benchmarks/transmon_eigen/results.toml` and `benchmarks/transmon_eigen/palace_crack/`.

Mode		geode-fem (GHz)	Palace (GHz)	Δ (%)
qubit	transmon qubit	3.452751	3.452749	0.00005
resonator	readout resonator	5.153	5.151	0.032
mode 2	box/cavity	15.465	15.461	0.029
port sheet	port-sheet artifact	17.490	17.490	0.001
mode 4	box/cavity	18.693	18.692	0.007
mode 5	box/cavity	20.703	20.698	0.026
mode 6	box/cavity	26.088	26.081	0.027

The 17.49 GHz mode is a spurious mode of the junction-sheet model, shared by both codes: Palace gives it a port EPR of 2.49×10^{-8}, and geode-fem's rank-1 shunt removes it.[2] The fixture is coarse: Palace's own refinements of it move the qubit from 3.4527 to 3.9528 GHz at $h_F/2$ (+14.5%) and to 4.1721 GHz at $p = 2$ on h_F (+20.8%), and the resonator by +6.1% at $h_F/2$, so the agreement above is a code-to-code comparison on one discretization.

5 The DeviceLayout Cross-Check

The published designs. The DeviceLayout.jl blog (Peairs and Carson, 2025) reports the initial design ($c = 620\,\mu$m, $L = 5000\,\mu$m) at $[4.14, 5.591]$ GHz and the final design ($c = 1217.678\,\mu$m, $L = 7243.955\,\mu$m) at $[3.005, 3.982]$ GHz, reached by PRIMA in nine Palace runs "within 1% of our target frequencies". The example source (AWS Center for Quantum Computing, 2025b) sets the targets to $[3.0, 4.0]$ GHz, works in the variables $x_1 = \sqrt{620\,\mu\text{m}/c}$ and $x_2 = 5000\,\mu\text{m}/L$, and stops when $\text{err} = \frac{1}{2}\sum_m (f_m - t_m)^2/t_m^2 < 10^{-4}$. Its frequencies are computed with Palace on DeviceLayout's curved mesh at $p = 2$, with a first-order absorbing far field and two 50 Ω readout ports; we call this *DeviceLayout's discretization*.

Reproduction at DeviceLayout's discretization. With the example's configuration file, Palace at DeviceLayout's discretization gives $[4.1683, 5.6132]$ GHz for the initial design and $[3.0164, 3.9956]$ GHz for the final one, 0.68% and 0.40% above the published initial pair and 0.38% and 0.34% above the final pair. The same cells with the PEC-box model used everywhere else differ from the configuration-file runs by at most 0.003%. The published runs' Palace version, mesher build and platform are not recorded (ours are in Section 11); every published value lies below ours, as a slightly coarser published mesh would make it, but which difference causes the 0.34–0.68% gap is not diagnosed.

The ladder and its rate. For each geometry, Palace runs at $p = 1$ on DeviceLayout's straight-sided mesh h and its uniform refinements $h/2$ and $h/4$ (family a, up to 12.6 million unknowns), and at $p = 2$ on h and $h/2$; geode-fem runs at $p = 1$ on h and $h/2$, on the same meshes. Every computed series rises under refinement (Figure 2); this is an observation, not a guarantee. We write the discretization error as $\propto h^r$ and call r the *convergence rate*. At the edge of a zero-thickness metal sheet the field behaves as $E \sim r_e^{-1/2}$ in the distance r_e from the edge (Meixner, 1972), so near the edge the eigenfunctions lie in H^s only for $s < 1/2$, and on uniform meshes the eigenvalue error decays as h^{2s} at any element order (Boffi, 2010): $r < 1$. The $p = 1$ ladders show pre-asymptotic rates of 0.61–0.62 (qubit) and 0.70–0.73 (resonator).

Extrapolation rule. The *estimate* is the mean of two extrapolations at rate $r = 1$ in the frequency, $2f_2 - f_1$: one from the $p = 1$ pair ($h/2$, $h/4$) and one from the $p = 2$ pair (h, $h/2$) (Figure 2, small open markers). $r = 1$ is the fastest rate the theory allows on uniform meshes, so the rule makes the smallest correction consistent with it, and its estimates lean low; for the $p = 2$ pair the rate is taken from theory, because two levels give no observed rate. The *band* spans six values, the same two pairs taken three ways (values for the initial qubit):

- at $r = 1$ in the frequency, the two values of the estimate (4.432 and 4.455 GHz);

[2] The junction patch has one free node whose hat-function gradient carries no port voltage but stores energy in S_Γ, with Rayleigh quotient $1/(L_J C_J)$, or $f_{LC} = 17.60$ GHz, lowered by the adjacent volume mass. This mechanism is our inference, supported by a synthetic test.

- at $r = 1$ in the eigenvalue variable $\lambda \propto f^2$, $\sqrt{2f_2^2 - f_1^2}$, which lie lower for a series that rises under refinement (4.415 and 4.451 GHz); the $p = 1$ value sets the low end, except on family-b resonators, where the $p = 2$ value does;
- at the observed rate of the $p = 1$ ladder's three levels (0.62 and 0.61 for the family-a qubits), borrowed by the $p = 2$ pair on the assumption that both element orders share the pre-asymptotic rate, which lie higher (4.677 and 4.582 GHz); the $p = 1$ value sets the high end.

The band is an extrapolation range, not a bound. For the initial qubit, the circuit-model cap of Section 6 trims it at 4.477 GHz.

The rule is Richardson extrapolation (Richardson, 1911) at the asymptotic rate rather than the observed one; Roache's grid convergence index (Roache, 1994) with a factor of 1.25, on the initial qubit's $p = 1$ ladder, is centred on 4.155 GHz with a half-width of 0.65 GHz, which reaches 4.81 GHz, above the circuit-model cap. As stated in Section 1, the rule was fixed after the static route had been computed; extrapolation at the observed rate alone puts the initial qubit at 4.63 GHz, above the cap, and the observed-rate values are kept only as the band's high side.

Table 3: Published frequencies, Palace at DeviceLayout's discretization (DL), the finest pass of Palace's adaptive run (a measured value, still rising), and the extrapolated estimates with their bands, in GHz, for the two refinement families of DeviceLayout's mesh h (a: gmsh's eight-way split; b: Palace's uniform refinement). Percentages are relative to the published value. Sources: `benchmarks/transmon_devicelayout_crosscheck/results.toml` (family a) and `benchmarks/transmon_devicelayout_controls/results.toml` (adaptive, family b).

			Adaptive,	Extrapolated estimate [band]	
	Published	DL	finest pass	Family a	Family b
Initial qubit	4.14	4.1683	4.4297 (+7.0%)	4.444 (+7.3%) [4.415, 4.677]*	4.472 (+8.0%) [4.458, 4.510]*
Initial resonator	5.591	5.6132	5.6986 (+1.9%)	5.740 (+2.7%) [5.723, 5.817]	5.776 (+3.3%) [5.768, 5.800]
Final qubit	3.005	3.0164	3.2468 (+8.0%)	3.265 (+8.7%) [3.236, 3.471]	3.292 (+9.6%) [3.278, 3.331]
Final resonator	3.982	3.9956	4.0552 (+1.8%)	4.088 (+2.7%) [4.077, 4.139]	4.113 (+3.3%) [4.106, 4.125]

*Under the circuit model of Section 6, the cap of 4.477 GHz trims the bands to [4.415, 4.477] and [4.458, 4.477].

Result. The qubit frequency extrapolates to 4.444 GHz on the initial design and 3.265 GHz on the final one, 7.3% and 8.7% above the published values; both resonator frequencies extrapolate to 2.7% above theirs (Table 3). Every published value lies below its band. With no extrapolation, the highest uniform levels, $p = 2$ on $h/2$ (7.0 million unknowns on the initial geometry; $p = 1$ on $h/4$ has more unknowns and lower values), already lie 4.2% and 4.7% above the published qubit frequencies. The estimates rest on Palace's $p = 1$ $h/4$ and $p = 2$ levels, which geode-fem did not run. On the levels it did run, geode-fem and Palace agree on the qubit to 10^{-4} % on both geometries. On the resonator they differ by 0.03% at h, where Palace adds 4,946 (initial) and 5,944 (final) elements by local bisection, and not at all in the six archived decimals at $h/2$, where Palace's logs report no bisection and the two codes differ only in the junction term (Section 3.3; the counts are in the logs). Three checks test the estimates; the adaptive runs check only the qubit.

A second refinement family. Family b refines the same h mesh with Palace's own uniform refinement instead of gmsh's split (family a). It first bisects where a sheet is one element wide (Section 4), so it is graded slightly toward the sheets, a plausible reason for its higher observed rates, 0.91 (initial) and 0.90 (final) against family a's 0.62 and 0.61. Its $h/2$ qubit is 3.9469 GHz against

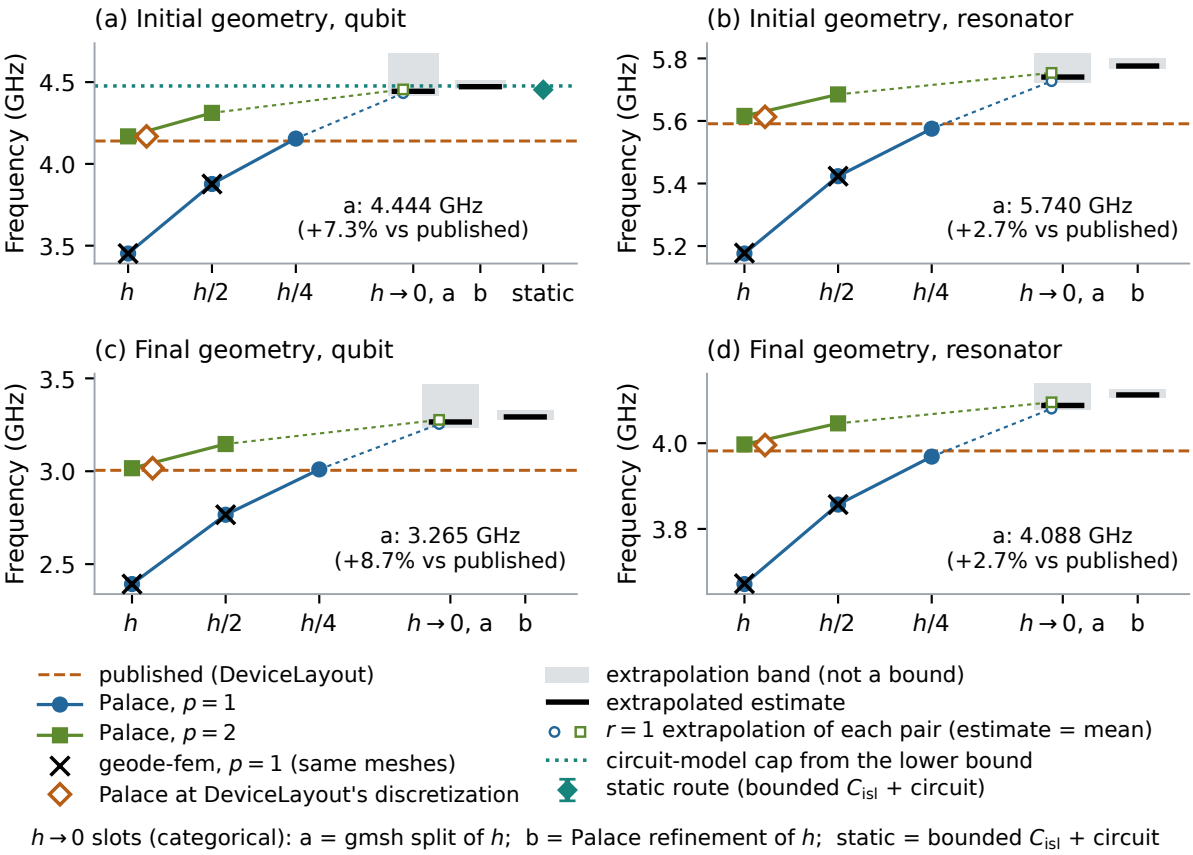


Figure 2: Resolution ladders of the two published DeviceLayout.jl SingleTransmon designs: Palace at $p = 1$ and $p = 2$ on DeviceLayout's mesh h and its family-a refinements, and geode-fem at $p = 1$ on the same meshes (crosses); the open diamond, just right of h, is Palace at DeviceLayout's own discretization, that of the published values (dashed). The "$h \to 0$" slots are categorical: the estimate with its band for the gmsh refinement family (a, labelled) and for Palace's refinement (b). The small open markers in slot a are the $r = 1$ extrapolations of the $p = 1$ pair ($h/2$, $h/4$) and the $p = 2$ pair (h, $h/2$), joined to the finer level of each pair; the estimate is their mean. In (a), the static route of Section 6 (natural box; the grounded bound lowers its low end by about 0.3 MHz) and the circuit-model cap, which trims the untrimmed bands drawn. Sources: `benchmarks/transmon_devicelayout_crosscheck/results.toml` and `benchmarks/transmon_devicelayout_controls/results.toml`.

gmsh's 3.8771 GHz, and at $h_F/2$ the two refinements of the fixture give 3.9528 and 3.8939 GHz, so the refinement scheme accounts for 1.5–1.8% of the $h/2$ level and the choice of starting mesh for only 0.15–0.43%. Its qubit estimates lie 0.64% and 0.83% above family a's, and every family-b band lies inside the family-a band (Table 3). We therefore quote the initial qubit frequency as 4.444–4.472 GHz, 7.3–8.0% above the published value, and the final one as 3.265–3.292 GHz, +8.7% to +9.6% over the published 3.005 GHz.

Adaptive refinement. A further check comes from Palace's adaptive mesh refinement, run on each geometry from its ladder's h mesh at $p = 2$. Each pass solves, estimates the error per element (summed over the qubit and resonator modes), and refines the elements that carry the largest 70% of it (Dörfler marking at 0.7), without keeping the mesh conformal (Figure 3 for the initial geometry). On the initial geometry, in five passes from 0.95 to 11.0 million unknowns, the qubit frequency rises from 4.1683 to 4.4297 GHz, 0.32% below the estimate, inside family a's band and below family b's;

the third pass, at 2.2 million unknowns, already exceeds the highest uniform level (4.3118 GHz at 7.0 million). On the final geometry, from 1.17 to 14.6 million unknowns, the qubit frequency rises from 3.0164 to 3.2468 GHz, 8.0% above the published 3.005 GHz, and the third pass again exceeds the highest uniform level (3.1884 against 3.1464 GHz). The finest final pass lies 0.56% and 1.39% below the two family estimates, again inside family a's band and below family b's. Both finest passes are measured values, not extrapolations, and neither run reached a plateau: each stopped when its last pass exceeded the run's size limit of 10^7 unknowns, not on its tolerance, and the last pass still moved the qubit by +0.55% (initial) and +0.63% (final). A three-point fit $f_\infty - CN^{-\alpha}$ through the last three passes gives 4.4552 GHz ($\alpha = 0.80$) and 3.2672 GHz ($\alpha = 0.83$), but +0.1% on the finest pass moves them by 24 and 16 MHz, so they are ill-conditioned estimates. The adaptive resonators rise more slowly and still lie below their bands, so the adaptive runs check the qubit estimates only.

A common-limit fit. The last check fits the $p = 1$ ($h/2$, $h/4$) and $p = 2$ (h, $h/2$) series of each ladder with one common limit and rate, exactly determined; it lies 0.5–1.3% above the rule's estimate on every first-family series, as expected of a rule that leans low, and +0.1% on the $p = 2$ $h/2$ level alone moves the initial-qubit fit by 20 MHz.

6 A Static Route: A Bounded Island Capacitance

The bound. For the reduced electrostatic problem (island at potential 1, ground and resonator at 0, feedline floating with zero net charge, and zero normal flux on the vacuum box), the potential ϕ satisfies $0 \le \phi \le 1$ by the maximum principle, and $C_{\text{isl}} = \int \varepsilon \nabla\phi \cdot \nabla\phi$ with $\varepsilon = \varepsilon_0\varepsilon_r$. Every conforming finite-element energy is an upper bound on C_{isl}, because C_{isl} is the minimum of the energy over admissible potentials. For the lower bound, take any piecewise polynomial field D on the mesh cut along the zero-thickness sheets. Integrating by parts and using $0 \le \phi \le 1$ on every term gives

$$\int D \cdot \nabla\phi \;\ge\; N(D) \;=\; Q_{\text{island}}(D) - |Q_{\text{feed}}(D)| - \|\operatorname{div} D\|_{L^1} - \|[D\cdot n]\|_{L^1(\mathcal{F})} - \|D\cdot n\|_{L^1(\mathcal{B})}, \quad (3)$$

where Q are the fluxes of D into the two conductors (for the exact field $D = \varepsilon\nabla\phi$, the negative of the physical displacement, they are the conductor charges), $[D\cdot n]$ is the normal jump on the interior faces $\mathcal{F}$ off the sheets, and $D\cdot n$ on the box faces $\mathcal{B}$ is the one-sided flux, which the zero-flux condition requires of the exact field but not of D. Cauchy–Schwarz, $\left(\int D\cdot\nabla\phi\right)^2 \le \int D\cdot\varepsilon^{-1}D \int \varepsilon\nabla\phi\cdot\nabla\phi$, then gives, whenever $N(D) > 0$,

$$C_{\text{isl}} \;\ge\; \frac{N(D)^2}{\int D\cdot\varepsilon^{-1}D}. \quad (4)$$

The bound holds for any D, so solver tolerance can only weaken it. Each L^1 norm is bounded above element by element and face by face, $\|p\|_{L^1(K)} \le |K|^{1/2}\|p\|_{L^2(K)}$, with $\|p\|_{L^2}^2$ integrated exactly by quadrature. We take D from the P2 potential, equilibrated patch by patch into Raviart–Thomas fluxes of order one to three, with an independent Crouzeix–Raviart flux as a check. The two bounds hold by theorem, for the exact arithmetic of the discretization on the polygonal geometry of the mesh, with lower bounds floored and upper bounds ceiled; the implementation is checked by tests, not formally verified. The tests compare it with an exact parallel plate and an analytic coaxial bracket, and mutation tests reject a grounded-feedline or jump-free lower bound. The bound is evaluated in plain f64, not interval arithmetic; the worst-case summation error is about 10^{-10} relative, far below the rounding. Refinement does not move the polygonal surfaces, so all meshes of one geometry bound the same exact capacitance, and a family bracket is the largest lower bound and the smallest upper bound.

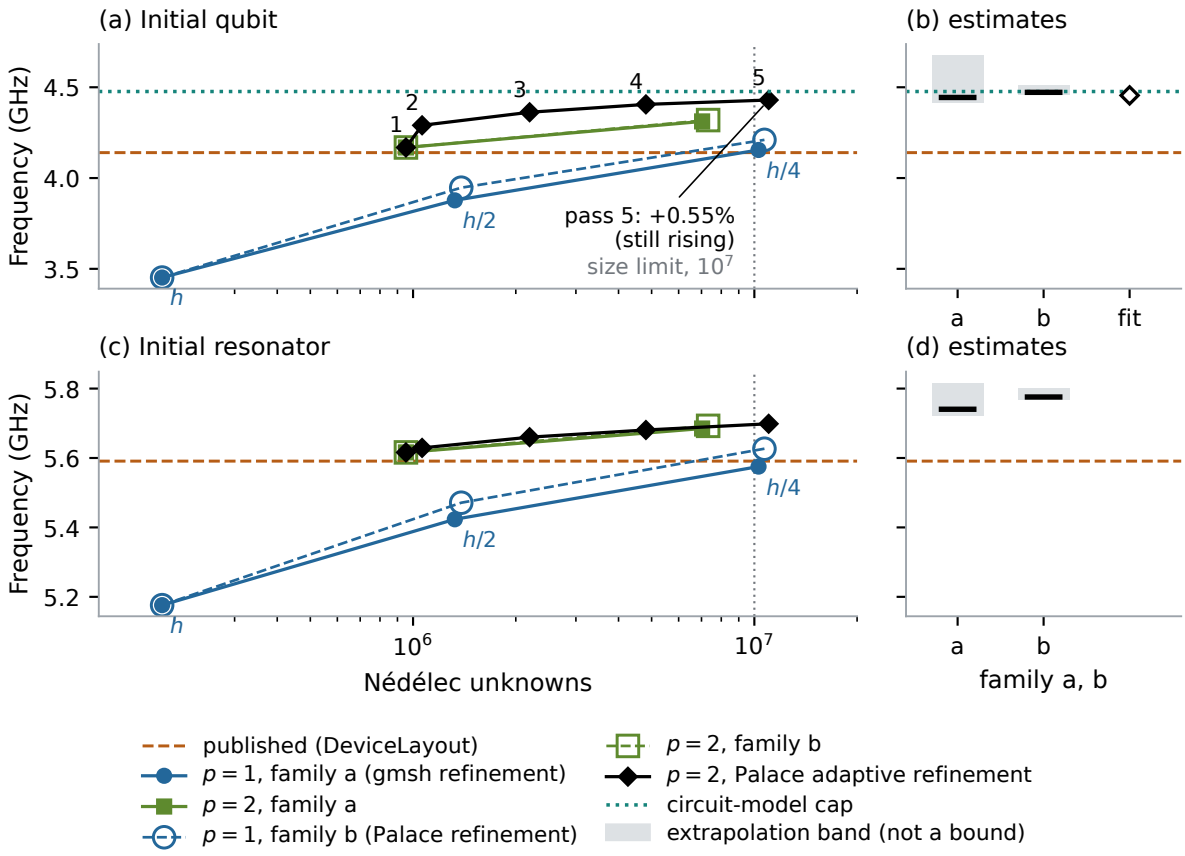


Figure 3: Checks of the extrapolated estimate on the initial geometry. (a, c) Every Palace level against the number of Nédélec unknowns, for the two refinement families of h (filled: gmsh's split, as in Figure 2; open: Palace's) and Palace adaptive refinement from h at $p = 2$ (diamonds, passes numbered in (a)); the dotted vertical line is the adaptive run's size limit, 10^7 unknowns. (b, d) The estimates of the two families with their bands and, for the qubit only, the adaptive run's ill-conditioned fit (the resonator fit is unreliable). In (a, b), the circuit-model cap. Sources: `benchmarks/transmon_devicelayout_crosscheck/results.toml` and `benchmarks/transmon_devicelayout_controls/results.toml`.

Two box models. The bound above puts no condition on the box walls; this is the *natural* box, in which no field line leaves the box. The eigenproblem instead makes the walls PEC (Section 3.2), the *grounded* box, with the walls held at ground potential. Constraining the potential on the box, to zero or to a floating constant, shrinks the set of admissible potentials, so it can only raise C_{isl}: the natural-box lower bound, and the cap below that uses only it, hold for the grounded box too. For an upper bound we re-solve the conforming P2 problems on the same meshes with the box grounded.

The unmodified device. Uniform refinement of the fixture with P1 and P2 elements gives the upper bounds of Figure 4a, from 137.71 to 83.02 fF (scalar) and from 136.69 to 85.01 fF (tensor), ceiled. On adaptively refined meshes the lower bound rises toward the falling upper bound (Figure 4b): the island capacitance of the unmodified initial design is bounded by $[77.40, 77.61]$ fF with the scalar $\bar{\varepsilon}$ and by $[79.56, 79.76]$ fF with the sapphire tensor. The bounded tensor-minus-scalar difference, computed from the unrounded bounds, is 1.95–2.35 fF, or 2.5–3.1% of the scalar value. With the box grounded, the tensor C_{isl} is bounded by $[79.56, 79.77]$ fF. On the finest mesh, grounding

the box raises the conforming value by 0.0065 fF, 8×10^{-5} relative, an estimate of the effect of the box model, whose size is bounded only by $[0, 0.21]$ fF. The admissible sets are nested, so the grounded bracket also covers a floating box. With C_J added, the transmon's $C_\Sigma = C_{\mathrm{isl}} + C_J$ is bounded by $[85.06, 85.27]$ fF for the eigenproblem's grounded box (tensor; $[85.06, 85.26]$ fF natural). The fixture itself overstates the tensor C_{isl} by at least 56.92 fF, which is at least 71.3% of the exact value, and the scalar one by at least 77.4%.

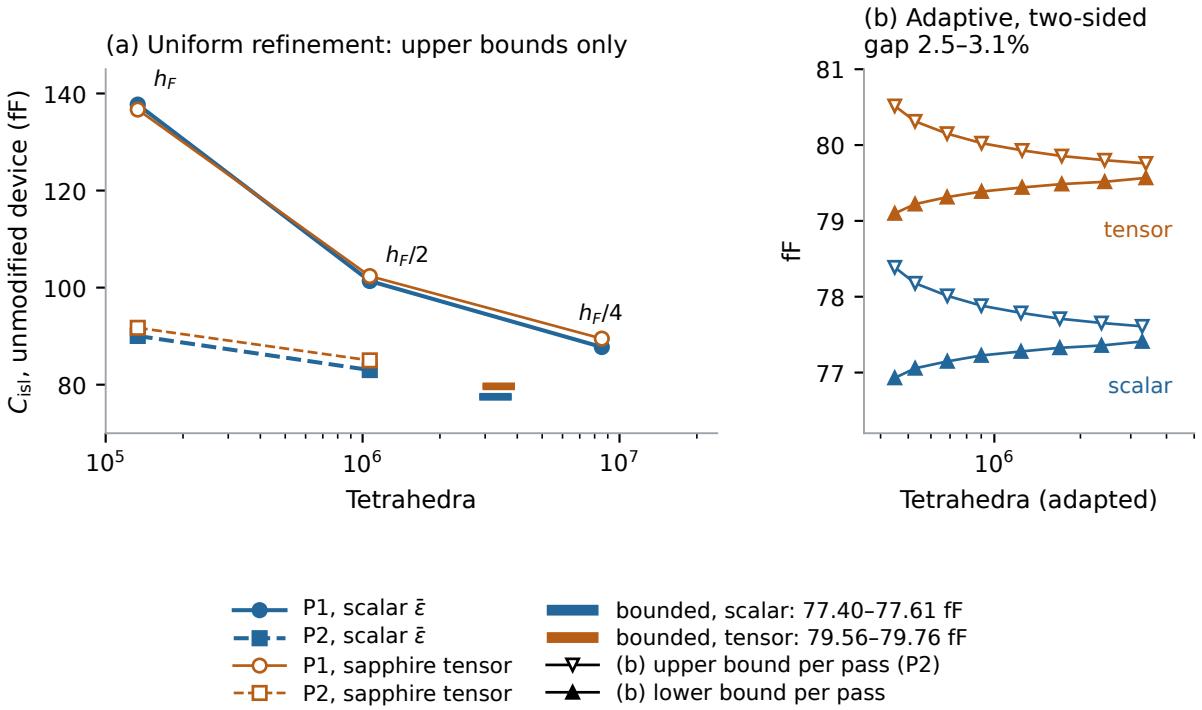


Figure 4: Mesh convergence of the island capacitance C_{isl} of the unmodified device, with the scalar trace-averaged permittivity (blue) and the sapphire tensor (orange). (a) Uniform refinement of the fixture (h_F, $h_F/2$, $h_F/4$): every marker is a conforming upper bound on the exact capacitance of the meshed polygonal geometry. The short bars are the two-sided bounds from adapted meshes, each drawn at the tetrahedron count of the mesh that sets its lower bound. (b) The adaptive passes behind those bars: on each adapted mesh the conforming P2 value (hollow, an upper bound) and the best equilibrated-flux lower bound (filled) close in on each other; the title gives the bounded tensor-minus-scalar gap. All values are for the natural box; the grounded box changes only the tensor upper bound, to 79.77 fF. Sources: `c_sigma_convergence.toml`, `c_sigma_bracket.toml` and `c_sigma_bracket_tensor.toml`.

From capacitance to the qubit frequency. Reading the grounded-box tensor bound through the two-node circuit of Section 3.3, over every corner of the C_{isl} bound and the L_s and k ranges, with the resonator at its extrapolated estimate, gives the initial dressed qubit in $[4.449, 4.458]$ GHz; the natural-box bound raises its low end by about 0.3 MHz. That range lies inside the eigenmode band of Section 5 and inside the range of the two family estimates, and the first family's estimate is 0.22% below its midpoint, a consistency check for the reason given in Section 1. The range is an estimate, because L_s, k and the resonator estimate are.

A model-dependent cap. Under the circuit model, a series inductance and a coupling to a higher mode can only lower the qubit, so the lower bound $C_{\mathrm{isl}} \geq 79.56$ fF, that is $C_\Sigma \geq 85.06$ fF,

caps it at $1/(2\pi\sqrt{L_J C_\Sigma}) = 4.477\,\mathrm{GHz}$. The cap is rigorous only given that model (one island node with $L_J \parallel C_J$, $L_s \geq 0$, coupling only to modes above the qubit, a quasi-static C_Σ) and rests on the fixture, not on the ladder meshes, which agree with it to 0.02% at h, so it is model-dependent and not in the Bounded tier. It does not depend on the extrapolation rule. Every computed level and both family estimates lie below it; the first family's common-limit fit, 4.4802 GHz, exceeds it by 3.6 MHz, within that fit's conditioning. With no extrapolation, the finest adaptive pass and the cap put the initial qubit between 4.4297 and 4.477 GHz, +7.0% to +8.1% over the published value; the lower end is a measured value, the upper end holds only under the circuit model.

Why the qubit is approached from below. The qubit mode is quasi-static; the quasi-static part of a first-order Nédélec field is the gradient of a P1 potential, which can only overstate the capacitance, so the computed qubit should approach its limit from below: on the fixture the P1 bound (136.69 fF, with C_J added) overstates C_Σ by a factor of 1.67, and the family estimates exceed the fixture's $p = 1$ qubit by 1.66–1.68 in f^2. This is a heuristic, not a proof: it omits L_s, the coupling, the rotational field and the distributed resonator.

7 Re-Optimization and Its Controls

At the extrapolated frequencies the published final design misses its targets. We repeat DeviceLayout's optimization against its own discretization, to compare optimizers in one environment, and against the extrapolated frequencies, to find the design that meets the targets. Cost is counted in *Palace runs*: one Palace eigensolve at DeviceLayout's discretization, about 4 min on 32 ranks. A Newton *step* places a new design and costs one Palace run; a *probe* is a Palace run that is not a step. geode-fem solves are not Palace runs, and their cost is given in seconds.

The capacitor_length derivative. DeviceLayout builds the island upward from the fixed junction, so a change of c stretches the strip that holds the island below its top and translates every node above that strip (Figure 1a). geode-fem turns this motion into a node velocity $\partial X/\partial c$, checks that every metal outline node in the stretched strip moves along its own edge (3,146 nodes on the initial mesh), and evaluates Eq. (1) for both modes in 0.44 s. On the initial mesh it matches a central difference of two re-solves of the moved mesh ($\pm 0.5\,\mu$m) to every printed digit: $-2.443227 \times 10^{-3}\,\mathrm{GHz}/\mu\mathrm{m}$ for the qubit and $-3.964 \times 10^{-6}\,\mathrm{GHz}/\mu\mathrm{m}$ for the resonator.

The total_length column. DeviceLayout re-lays the meander when L changes, so no fixed-topology node motion represents it. A transmission-line model holds $f_r(L + \delta)$ constant and takes $\partial f_q/\partial L = 0$, with $\delta = 216\,\mu$m from one finite difference of two geode-fem $p = 1$ solves ($L = 5000$ and $5250\,\mu$m). δ is weakly determined: remesh noise of about 0.15% in either f_r moves it by about $\pm 175\,\mu$m. The model enters only the Jacobian, so a poorer Jacobian costs steps, not accuracy.

Newton on DeviceLayout's objective. DeviceLayout re-meshes every design, so the $p = 1$ objective on h is not smooth in the parameters: from $c = 620$ to $650\,\mu$m the remeshed $p = 1$ difference of the qubit is $-5.23 \times 10^{-3}\,\mathrm{GHz}/\mu\mathrm{m}$ against a fixed-topology derivative of -2.28×10^{-3} averaged over the interval (the mean of its end values), because remeshing alone moves the qubit by about 2.6%, more than the 2.0% the step moves it. The iteration therefore runs Newton directly on the objective F, the frequencies at DeviceLayout's discretization, in DeviceLayout's variables (x_1, x_2), in which the frequencies are close to linear and nearly decoupled: a probe in either variable moves its own mode by 4.5% and the other by at most 0.05%. The c column of the Jacobian is geode-fem's $p = 1$ derivative on h, $\partial f_h/\partial c$, scaled by the ratio $S = F/f_h$ of the two discretizations

at the same design. Each step costs one geode-fem $p = 1$ solve with its gradient (19–25 s) and one Palace run; the geode-fem solves of the one-time δ calibration are not Palace runs.

Table 4: Palace runs to DeviceLayout's criterion (err $< 10^{-4}$) at its discretization, all in our Palace environment except the published PRIMA run. Every Palace run a method uses counts, with the initial design as run 1 and probes included, among them the probe that sets the Palace δ (Broyden's second probe, shared by the two rows); geode-fem solves are not counted. Each count is one trajectory. PRIMA runs at `rhobeg` $= 0.2$, its initial trust-region radius, unless stated; the Newton variants use the transmission-line L column. Sources: the `optimize` and `controls` artifacts under `benchmarks/transmon_devicelayout_*/`.

Method	c column	δ from	Palace runs	err after first step	Design at criterion (μm)
PRIMA, published	none	none	9	unpublished	(1217.678, 7243.955)
PRIMA, our environment	none	none	10	n/a	(1209.125, 7207.551)
`rhobeg` 0.1	none	none	10	n/a	(1235.705, 7216.766)
`rhobeg` $(x_u - x_l)/2$	none	none	12	n/a	(1245.900, 7291.401)
Newton	shape derivative	geode-fem	3	7.9×10^{-4}	(1225.346, 7223.250)
Newton	scaling law	geode-fem	3	1.7×10^{-4}	(1232.664, 7221.270)
Newton	scaling law	Palace	4	2.1×10^{-4}	(1229.026, 7225.712)
Broyden	two probes	none	5	6.2×10^{-4}	(1224.847, 7232.447)

Against DeviceLayout's discretization. Table 4 and Figure 5(b) compare the optimizers on one objective. Newton with the shape-derivative column meets the criterion at its second step, after 3 Palace runs, and its three geode-fem solves add about 70 s, against about 4 min per Palace run; a third step reaches err $= 1.4 \times 10^{-7}$ at $(1236.360, 7224.900)\,\mu$m. Two gradient-free controls test where the 3 runs come from. The first replaces the c column by the scaling law $\partial F_q/\partial x_1 = F_q/x_1$, which follows from $f_q \propto 1/\sqrt{C_\Sigma}$ with C_Σ taken proportional to c. With the geode-fem δ it also takes 3 runs, and its first step has an err 4.7 times lower (Table 4), about 2.2 times closer in frequency. With $\delta = 563.879\,\mu$m taken instead from a Palace probe (c fixed, L from 5000 to 5263 μm), so that no geode-fem input remains, it takes 4 runs: the same two steps, plus the probe. This δ is 2.6 times the geode-fem one, more than the $\pm 175\,\mu$m noise estimate, and costs no extra step. The second control, Broyden's method (Broyden, 1965) from two forward finite-difference probes, takes 5.

PRIMA in our environment. DeviceLayout's own PRIMA call, re-run verbatim in our environment, takes 10: runs 1–5 are its initial interpolation set of $2n + 1 = 5$ points (Powell, 2009), and run 9 misses the criterion by 15% (err $= 1.150 \times 10^{-4}$). In this environment the published final design already meets the criterion (err $= 1.5 \times 10^{-5}$), so the environment moves PRIMA's trajectory, not the attainability of the target; the published count of 9 is from DeviceLayout's environment. With `rhobeg` at 0.1 instead of 0.2, PRIMA also takes 10 runs, and at the largest radius BOBYQA accepts on DeviceLayout's box, $(x_u - x_l)/2$, it takes 12. Each setting is a single run. In one environment, then, Newton's method in DeviceLayout's variables takes 3 or 4 Palace runs with any of the three Jacobians and PRIMA 10 to 12; the one run between 3 and 4 is the Palace probe that geode-fem's δ calibration replaces.

Where the first steps land. The shape-derivative and Broyden first steps land at $c = 1338.338$ and $1320.522\,\mu$m, past about $1230\,\mu$m at the criterion: both follow a local slope, $\partial F_q/\partial x_1 = 3.66$ and 3.71 GHz, within about 1.5% of each other and 11–12% shallower than the scaling law's $F_q/x_1 = 4.168$ GHz.

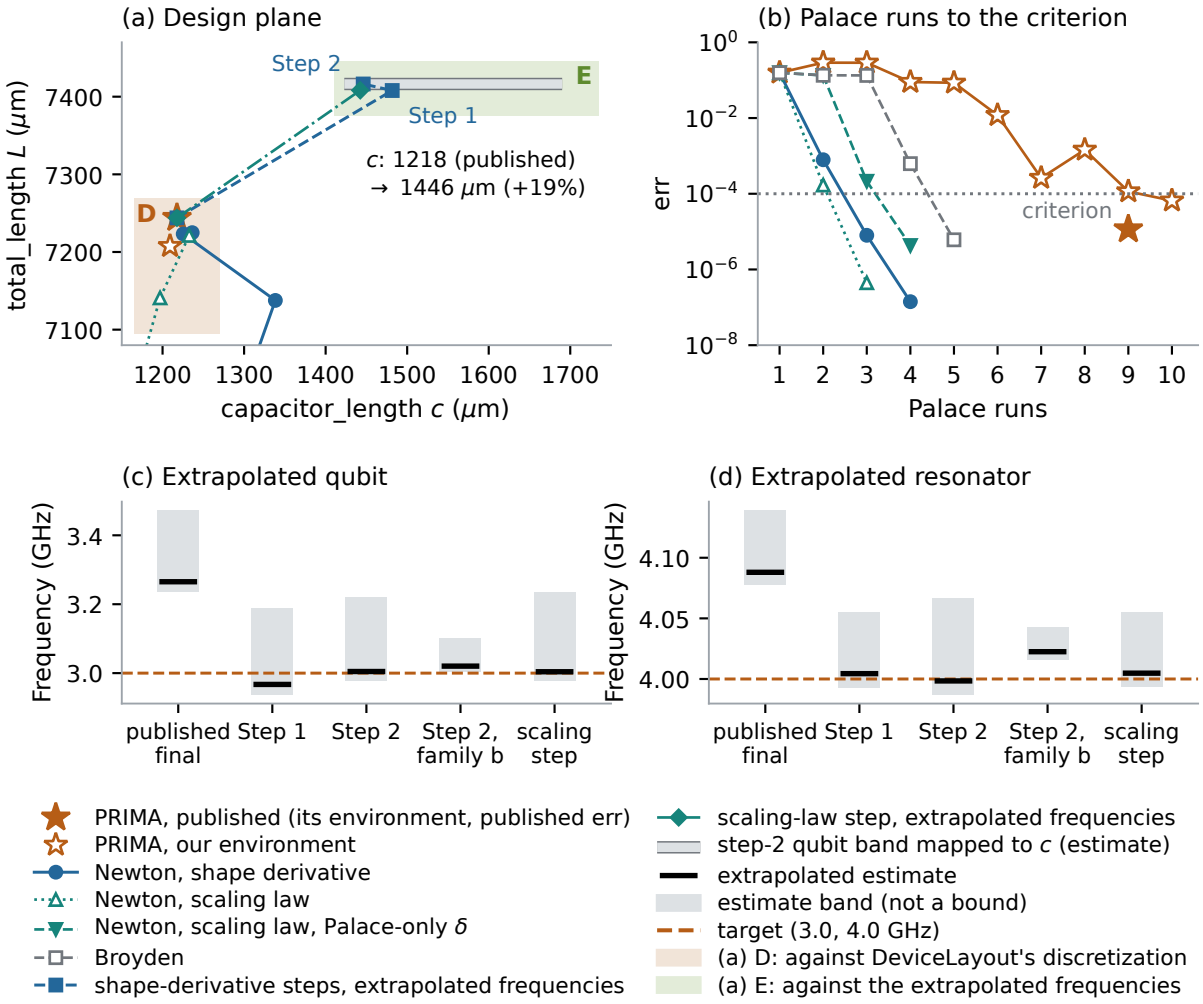


Figure 5: The re-optimization and its controls. (a) The design plane near the optimum, in two regions. D: the published final design, PRIMA's end point in our environment and the Newton iterates against DeviceLayout's discretization (the first legs start at the initial design, off the plot). E: the steps against the extrapolated frequencies, with the step-2 qubit band mapped linearly to capacitor_length (an estimate); the text gives the published and re-optimized capacitor lengths. (b) DeviceLayout's err against Palace runs at its discretization in our environment, probes counted (the Palace-δ Newton's run 2 is the L probe it shares with Broyden), PRIMA at `rhobeg` $= 0.2$; the published PRIMA point is at its 9 runs and the err of its published frequencies, 1.151×10^{-5} (DeviceLayout's environment). This counts Palace runs, not total cost: each shape-derivative Newton point also costs one geode-fem solve. (c, d) Extrapolated estimates and bands against the targets, with step 2 in both refinement families. Sources: the `optimize`, `controls` and `crosscheck` artifacts under `benchmarks/transmon_devicelayout_*/`.

Against the extrapolated frequencies. Starting from the published final design, each Newton step uses the extrapolated estimate and the geode-fem derivative at the current design, and each new design is evaluated by a full ladder with the rule of Section 5 (Table 5). The first overshoots the qubit by -1.09%; the second lands at $+0.16\%$ and -0.04% of the targets, with both targets inside the bands. On both ladders geode-fem and Palace agree on the qubit to $2 \times 10^{-4}\,\%$. One scaling-law step from the same start, evaluated by the Palace half of the ladder, lands 3.6 and 8.2 μm short of step 2 and at $+0.13\%$ and $+0.12\%$ of the targets: one gradient-free step does what the two shape-derivative steps did.

Table 5: Steps against the extrapolated frequencies, in GHz. Each design is evaluated by a full resolution ladder (the scaling-law step and the second family by Palace alone) and the rule of Section 5; values are estimates with their bands. Family b refines step 2's h mesh with Palace's uniform refinement instead of gmsh's. Sources: the `optimize`, `controls` and (published final) `crosscheck` artifacts under `benchmarks/transmon_devicelayout_*/`.

Design	c (μm)	L (μm)	f_q [band]	f_r [band]
Published final	1217.678	7243.955	3.265 [3.236, 3.471]	4.088 [4.077, 4.139]
Step 1	1481.657	7407.945	2.967 [2.938, 3.188]	4.004 [3.993, 4.055]
Step 2	1446.078	7416.352	3.005 [2.979, 3.221]	3.998 [3.987, 4.067]
Step 2, family b	1446.078	7416.352	3.020 [3.003, 3.101]	4.022 [4.016, 4.043]
Scaling-law step	1442.479	7408.181	3.004 [2.977, 3.235]	4.005 [3.994, 4.055]

The re-optimized design. Step 2 is the design whose extrapolated frequencies sit on the targets: $c = 1446.078\,\mu$m, 18.8% longer than published, with a resonator 2.4% longer ($+172\,\mu$m). The second refinement family at step 2 gives 3.0203 [3.0033, 3.1014] GHz for the qubit and 4.0225 GHz for the resonator, 0.52% and 0.60% above the first family, the same spread as on the initial geometry; both targets lie just below its band, so on family b step 2 is slightly short. Mapped to 3.0 GHz by the scaling law $f_q \propto c^{-1/2}$, the two families' qubit estimates call for $c = 1451$ and $1466\,\mu$m, +19.1% and +20.4% over the published length. Their bands map to +17% to +37% (family a) and +19% to +27% (family b) under the scaling law, and to +17% to +39% (c from about 1423 to $1690\,\mu$m) and +19% to +28% linearly through $dF/dc \approx S\,\partial f_h/\partial c = -9.05 \times 10^{-4}$ GHz/μm at the step-1 design, centred on the target. Under either mapping the first range contains the second. Since the estimates lean low, the common-limit fit, which puts step 2 at 3.039 and 4.019 GHz, would call for a longer capacitor still.

8 How geode-fem Was Designed and Built

geode-fem realizes the whiteroom operator specification (Dunnavant, 2026), whose four layers, L1–L4, restate Palace's cited source as pure functions, unfold those into compositions of base algebraic primitives, recast them where possible as global tensor operations, and express them in a small graph-evaluation calculus. Author contributions: C.D. wrote that specification, gave design input and review on the solver's operator layer, and contributed code: the migration to Rust 2024, the Metal (Apple GPU) backend, the present layout of the core crate and the examples, and the geode-util crate. R.W. ran the research and the AI agents that produced the rest of the code, the experiments and the text of this paper, made every scoping and claim decision, and is responsible for the results. The agents were AI models working in separate roles: Anthropic's Claude (Opus, Fable and Sonnet models, through Claude Code) and OpenAI's GPT Sol and GPT Astra models (via Codex). They were coordinated by Loom (Walters, 2025), an issue, worktree and pull-request workflow, and by Squad, a message room the agents share. Nearly every change was reviewed by a separate agent session. The agent reviews of this paper recomputed its numbers from the archived artifacts, reproduced runs, and mutation-tested the scripts that check the text and figures against them.

9 Discussion

What is established, and how. Table 6 sorts the results by the kind of evidence behind them. The brackets are the only bounds, and the cap follows from a bracket only under a circuit model.

Table 6: Evidence tiers. Bounded: a two-sided bound that holds by theorem for the meshed polygonal geometry. Model-dependent: a rigorous consequence of a bound under a stated circuit model. Measured: a solve on a stated discretization. Estimated: an extrapolation with a stated rule. Consistency: an agreement that is not independent evidence.

Tier	Results
Bounded	Island capacitance C_{isl} of the unmodified device, without C_J, natural box: $[77.40, 77.61]$ fF (scalar), $[79.56, 79.76]$ fF (tensor); grounded box, the eigenproblem's model: $[79.56, 79.77]$ fF (tensor), so $C_\Sigma = C_{\mathrm{isl}} + C_J$ in $[85.06, 85.27]$ fF and $e^2/2hC_\Sigma$ in $[0.2271, 0.2278]$ GHz; tensor-minus-scalar gap 2.5–3.1%
Model-dependent	The initial-qubit cap of $4.477\,\mathrm{GHz} = 1/(2\pi\sqrt{L_J C_\Sigma})$ at the lower bound on C_Σ: two-node circuit, quasi-static C_Σ, bounded on the fixture's polygonal geometry, not on the ladder meshes; with the finest adaptive pass, $[4.4297, 4.477]$ GHz; the same $e^2/2hC_\Sigma$ read as the transmon's charging energy E_C/h
Measured	Same-mesh agreement (Table 2); the reproduction within 0.7%; every ladder level and adaptive pass (adaptive qubit frequencies +7.0% and +8.0%, still rising); the gradient check; Palace runs to DeviceLayout's criterion, one trajectory each (Table 4)
Estimated	Extrapolated frequencies and bands (Table 3); the adaptive and common-limit fits; the static-route qubit range; L_s and k; the box-model effect on C_{isl}, 8×10^{-5}; the re-optimized capacitor, +19 to +20% over two families and about +17 to +39% within the band
Consistency	The two routes agree to 0.22% on the initial qubit; the extrapolation rule was fixed after the static route was known, so this is not a blind prediction
Not claimed	Agreement with measurement; a verified hit of 3.0/4.0 GHz in the $h \to 0$ limit; a global optimum; an advantage of the shape derivative on this problem; eigenmode-participation (EPR) derivatives, which need eigenvector derivatives (Nelson, 1976); a speed advantage over Palace

Where the shift comes from. The evidence places the 7–8% shift in the discretization, not in arithmetic precision or in the eigensolver. geode-fem computes in double precision (f64), and Palace is the build pinned in `reference/palace/docker/Dockerfile`, which sets no single-precision option. On shared meshes the two codes agree on the qubit frequency to 2×10^{-4}% (Sections 5 and 7); a loss of precision large enough to matter in either code would appear as a same-mesh disagreement far above that level. Palace is used here as a frequency-domain finite-element eigensolver, not a time-domain (FDTD) code: it returns the eigenmodes of the discrete problem that its input mesh and element order define, and given the same mesh, geode-fem, with its own assembly and eigensolver, returns the same modes (Table 2; the crosses in Figure 2). The error lies in that input: refining the mesh from h to $h/2$ moves both codes' frequencies up together (Figure 2), and the published values come from one discretization.

Why the shape derivative gives no advantage here. Three properties of this benchmark explain why Newton's method needs 3 or 4 Palace runs whatever its Jacobian. First, the problem has two nearly decoupled parameters, close to linear in DeviceLayout's variables, so Newton's method converges with any reasonable Jacobian: the shape derivative, a two-probe secant, or the scaling law $f_q \propto x_1$, which costs nothing. Second, a finite-difference gradient costs only $N + 1 = 3$ solves. Third, the coarse-mesh, fixed-topology derivative is the slope of a different discrete problem from the one the iteration evaluates (DeviceLayout remeshes every design and Palace evaluates it on a curved mesh at $p = 2$); once scaled by the ratio S it gives about what Broyden's probe gives (Section 7). That the derivative is exact for its own discrete problem and cheap (0.44 s) does not change the number of Palace runs.

Derivatives pay off in other regimes. With N parameters, a discrete adjoint gives all N derivatives of one objective for about one extra solve, where finite differences need N more; coupled targets with

no per-parameter scaling law need a computed Jacobian; and freeform shapes have many parameters by construction. A companion paper on adjoint shape optimization of a conformal patch antenna (Walters, 2026) measures this on the worst-of-band reflection $|S_{11}|$ over 2.400–2.4835 GHz (artifact: `benchmarks/patch_antenna_conformal/tuned/tuned_summary.toml`). At 38 shape parameters the adjoint reaches a -9.5 dB target with about 11 times fewer solves than the fastest gradient-free run that reached it, a tuned full-covariance CMA-ES comes within 7.7 times at -9 dB, and runs that exhausted their budget before reaching the -9.5 dB target bound the ratio from below at 2.2–3; at 4 parameters the ratio is 1.1 at -9 dB and 2.4 at -9.5 dB.

Threats and limitations. Agreement between two codes is evidence against implementation error, not against shared modelling choices such as the port sheet behind the 17.49 GHz mode. The extrapolations assume the rate the sheet-edge singularity allows on uniform meshes, their high sides are wide, neither adaptive run reached a plateau, and L_s and k were extrapolated from two or three discretizations. Every limit $h \to 0$ here is a limit of DeviceLayout's model (zero-thickness PEC sheets, lossless and linear, open readout ports in the ladders); film thickness and kinetic inductance remove the edge singularity and move the frequencies by amounts not measured here. The element edges along the sheet outlines are mostly 10–12 μm long on h and so 2.5–3 μm on $h/4$, more than ten times a 0.2 μm film, on every uniform mesh. Palace's adaptive runs refine at most four times: a typical outline edge stays above about 0.6 μm, but the 1.0–1.8 μm edges of the junction leads, where the qubit's error indicator is largest, could reach 0.06–0.11 μm, comparable to or below the film (an estimate: no committed record holds the adaptive meshes' minimum edge). There the zero-thickness model, which every mesh discretizes, is an approximation whose effect is not quantified; the shift measured here is the discretization error within that model, for one device.

What the shift means for a design. Table 7 reads the initial qubit as a total capacitance. The bound gives $E_C/h = e^2/2hC_\Sigma$ in $[0.2271, 0.2278]$ GHz for the grounded box, Bounded as a function of C_Σ and model-dependent as the transmon's charging energy (a single-island lumped picture). Under the two-node circuit with the L_s and k ranges, the published 4.14 GHz implies a C_Σ 15.5–16.0% above the bound, so E_C and the anharmonicity, close to $-E_C$ (Koch et al., 2007), are larger than the published discretization implies by as much. The bare LC reading at fixed L_J omits L_s and k, so it reads every frequency as a larger C_Σ, 16.6–17.0% above the bound for the published value (rounded outward). Of that excess, taken as a factor, the measured finest adaptive pass accounts for 1.145, and the remaining 1.9–2.1% splits between the still-rising pass and L_s and k. The transition f_{01} rises by a little less than the linear mode, about 6.6% at the finest adaptive pass, and L_J can retune the qubit frequency but not E_C.

Table 7: The initial qubit read as a total capacitance C_Σ, with $E_C/h = e^2/2hC_\Sigma$; "bare" is the LC reading at fixed L_J ($L_s = k = 0$), "two-node" the circuit of Section 3.3 with the L_s and k ranges, given first for the published value; bare rows omit L_s and k, so their C_Σ lies above the (grounded-box) bound. Sources: `c_sigma_bracket_grounded_box.toml`, Table 3 and `two_node_reading.toml` in the paper's directory (the `transmon_loop_inductance` anchor uses another topology).

Source	Tier	f_q (GHz)	C_Σ (fF)	E_C/h (GHz)
Bound	Bounded	–	$[85.06, 85.27]$	$[0.2271, 0.2278]$
Static route	Estimated	$[4.449, 4.458]$	85.8–86.1 (bare)	0.225–0.226 (bare)
Adaptive finest pass	Measured	4.4297	86.87 (bare)	0.223 (bare)
Published	–	4.14	$[98.51, 98.64]$; 99.45 (bare)	0.196–0.197; 0.195 (bare)

What a practitioner can do. The standard remedy for the slow edge rate is to refine toward the edges, with a mesh graded toward the edges or with *hp* refinement (Demkowicz, 2006): Palace's adaptive runs (Section 5) and the gap-refined, fourth-order setting of Ye et al. (2025), within 0.3% of their extrapolated values, are instances. A first check for a copied layout is the strip width: on the fixture, 993 of the 1,007 edges that trigger Palace's bisection lie where a resonator or feedline strip is one element wide (Section 4). Surface-loss estimates (Martinis, 2022; Wang et al., 2015) need the same edge resolution as the frequencies.

10 Conclusion

Palace reproduces the published DeviceLayout.jl SingleTransmon frequencies at the example's own discretization, and that discretization reads the qubit frequencies low: adaptive runs put them 7.0% and 8.0% above their published values with no extrapolation, and the estimates of two refinement families, which lean low, put them 7.3–9.6% above. geode-fem reproduces Palace's qubit on every level both codes ran, and a two-sided bracket on the island capacitance, read through a circuit model, is consistent with that shift and caps the initial qubit under that model, a consistency check rather than a blind prediction. At the extrapolated frequencies the 3.0/4.0 GHz design needs a capacitor about 19–20% longer than published. Against the example's own objective, Newton's method in DeviceLayout's variables reaches the criterion in 3 or 4 Palace runs with any of three Jacobians, against 10 to 12 for PRIMA; geode-fem's shape derivative gives no advantage here. For design examples that are meant to be copied, the discretization is part of the result and belongs in the report beside it.

11 Reproducibility

Every result is read from an archived, scripted artifact.[3] Each caption names its artifact; the Palace cells are under `reference/fixtures/transmon_palace/`, and the DeviceLayout meshes are generated by `reference/julia/generate_transmon_param.jl`. The versions are geode-fem commit `3174015` for the same-mesh agreement, with the commit of each later result recorded in its artifact, Palace `fba6a5b`, DeviceLayout.jl v1.15.0, PRIMA.jl 0.2.4 and Burn 0.21. Palace ran on AWS Graviton4 hosts, and the capacitance brackets on a 28-core workstation and on Graviton4. geode-fem's direct solves at $h/2$ peaked at 94 and 136 GB, so only Palace ran larger cells.

Palace's ladder cells use SLEPc shift-invert at tolerance 10^{-8}, `Target` 3.0 (initial) or 2.0 GHz (final), and its default direct solver, SuperLU ($p = 1$ on h), or AMS-preconditioned GMRES at relative tolerance 10^{-9}, on 16 ($p = 1$ on h) or 32 MPI ranks; the fixture ran on 8, and the runs at DeviceLayout's discretization keep the example's tolerances (10^{-6} and 10^{-7}, `Target` 2.5 GHz). geode-fem's Lanczos shifts are 5 and 4 GHz, with relative residuals below 4×10^{-8}. The h, $h/2$ and $h/4$ meshes have 133,355, 1,066,840 and 8,534,720 tetrahedra (initial) and 163,431, 1,307,448 and 10,459,584 (final). The meshes were made with gmsh 4.15.2 (`gmsh_jll`) under Julia 1.10.11, and Palace was built against the MPICH of the Rocky Linux 9 image in `reference/palace/docker/Dockerfile`.

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[3] The geode-fem source is public at https://github.com/rjwalters/geode-fem. Archival DOI to be added at deposit.

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